

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
2.642	150.0	145.0	295.0	7693.676	26.08

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[242,4]	[207,16]	38.971	BRANCH	BRANCH
2	[245,2]	[242,4]	2.414	TIP	BRANCH
3	[243,7]	[242,4]	2.0	TIP	BRANCH
4	[137,10]	[80,52]	96.426	BRANCH	BRANCH
5	[137,10]	[207,16]	86.012	BRANCH	BRANCH
6	[133,14]	[137,10]	5.243	TIP	BRANCH
7	[3,31]	[52,15]	70.627	TIP	BRANCH
8	[52,15]	[53,22]	6.414	BRANCH	BRANCH
9	[68,3]	[52,15]	19.556	TIP	BRANCH
10	[207,23]	[207,16]	6.828	TIP	BRANCH
11	[347,1]	[345,16]	14.828	TIP	BRANCH
12	[345,16]	[298,34]	53.042	BRANCH	BRANCH
13	[345,16]	[368,28]	27.142	BRANCH	BRANCH
14	[423,21]	[469,24]	51.213	BRANCH	BRANCH
15	[423,21]	[418,55]	35.071	BRANCH	BRANCH
16	[388,15]	[423,21]	43.042	TIP	BRANCH
17	[36,38]	[53,22]	24.385	TIP	BRANCH
18	[53,22]	[61,39]	18.899	BRANCH	BRANCH
19	[469,24]	[487,81]	69.314	BRANCH	BRANCH
20	[469,24]	[438,60]	47.426	BRANCH	BRANCH
21	[382,38]	[368,28]	18.899	TIP	BRANCH
22	[368,28]	[364,67]	45.042	BRANCH	BRANCH
23	[298,34]	[301,45]	55.627	BRANCH	BRANCH
24	[298,34]	[301,45]	10.828	BRANCH	BRANCH
25	[61,39]	[80,52]	23.385	BRANCH	BRANCH
26	[61,39]	[37,62]	33.698	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[243,22]	[238,46]	25.071	TIP	BRANCH
28	[259,83]	[238,46]	45.87	TIP	BRANCH
29	[235,46]	[238,46]	2.0	TIP	BRANCH
30	[80,52]	[80,58]	5.0	BRANCH	BRANCH
31	[20,53]	[7,56]	13.243	BRANCH	BRANCH
32	[21,47]	[20,53]	5.0	TIP	BRANCH
33	[20,53]	[37,62]	19.314	BRANCH	BRANCH
34	[418,55]	[438,60]	21.071	BRANCH	BRANCH
35	[418,55]	[394,61]	25.485	BRANCH	BRANCH
36	[3,50]	[7,56]	6.657	TIP	BRANCH
37	[1,68]	[7,56]	13.485	TIP	BRANCH
38	[151,78]	[80,58]	83.012	TIP	BRANCH
39	[80,58]	[46,85]	47.113	BRANCH	BRANCH
40	[438,60]	[448,81]	24.142	BRANCH	BRANCH
41	[384,47]	[394,61]	17.728	TIP	BRANCH
42	[394,61]	[394,64]	2.0	BRANCH	BRANCH
43	[37,62]	[46,85]	25.728	BRANCH	BRANCH
44	[394,64]	[366,67]	28.243	BRANCH	BRANCH
45	[394,64]	[402,84]	21.899	BRANCH	BRANCH
46	[364,67]	[366,67]	1.0	BRANCH	BRANCH
47	[339,74]	[364,67]	26.899	TIP	BRANCH
48	[366,67]	[385,99]	39.456	BRANCH	BRANCH
49	[448,81]	[452,82]	3.414	BRANCH	BRANCH
50	[448,81]	[407,85]	42.899	BRANCH	BRANCH
51	[487,81]	[452,82]	34.0	BRANCH	BRANCH
52	[487,81]	[481,145]	67.899	BRANCH	BRANCH
53	[452,82]	[435,129]	52.627	BRANCH	BRANCH
54	[402,84]	[407,85]	4.414	BRANCH	BRANCH
55	[402,84]	[385,99]	22.799	BRANCH	BRANCH
56	[46,85]	[16,116]	44.77	BRANCH	BRANCH
57	[407,85]	[435,129]	55.184	BRANCH	BRANCH
58	[385,99]	[385,105]	5.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[385,105]	[362,119]	31.314	BRANCH	BRANCH
60	[385,105]	[387,120]	14.414	BRANCH	BRANCH
61	[5,105]	[12,116]	12.899	TIP	BRANCH
62	[12,116]	[16,116]	3.0	BRANCH	BRANCH
63	[1,130]	[12,116]	17.142	TIP	BRANCH
64	[16,116]	[59,147]	54.426	BRANCH	BRANCH
65	[306,117]	[362,119]	60.799	TIP	BRANCH
66	[362,119]	[370,130]	12.899	BRANCH	BRANCH
67	[387,120]	[426,137]	45.042	BRANCH	BRANCH
68	[387,120]	[370,130]	20.314	BRANCH	BRANCH
69	[273,127]	[267,129]	5.828	TIP	BRANCH
70	[267,129]	[267,138]	8.0	BRANCH	BRANCH
71	[223,129]	[267,129]	46.728	TIP	BRANCH
72	[435,129]	[434,137]	7.414	BRANCH	BRANCH
73	[370,130]	[342,160]	43.527	BRANCH	BRANCH
74	[426,137]	[432,137]	5.0	BRANCH	BRANCH
75	[426,137]	[403,171]	43.87	BRANCH	BRANCH
76	[432,137]	[434,137]	1.0	BRANCH	BRANCH
77	[432,137]	[432,184]	46.0	BRANCH	BRANCH
78	[434,137]	[481,145]	49.485	BRANCH	BRANCH
79	[222,170]	[267,138]	58.426	TIP	BRANCH
80	[267,138]	[342,160]	86.598	BRANCH	BRANCH
81	[481,145]	[479,175]	29.828	BRANCH	BRANCH
82	[89,129]	[59,147]	36.456	TIP	BRANCH
83	[54,178]	[59,147]	32.071	TIP	BRANCH
84	[110,156]	[126,160]	17.485	TIP	BRANCH
85	[128,164]	[126,160]	3.828	TIP	BRANCH
86	[144,151]	[126,160]	20.314	TIP	BRANCH
87	[342,160]	[342,164]	3.0	BRANCH	BRANCH
88	[342,164]	[333,207]	46.314	BRANCH	BRANCH
89	[342,164]	[369,187]	35.113	BRANCH	BRANCH
90	[403,171]	[384,177]	20.485	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[403,171]	[432,184]	32.971	BRANCH	BRANCH
92	[479,175]	[442,189]	41.799	BRANCH	BRANCH
93	[479,175]	[495,214]	44.213	BRANCH	BRANCH
94	[382,177]	[384,177]	1.0	BRANCH	BRANCH
95	[378,172]	[382,177]	5.243	TIP	BRANCH
96	[382,177]	[369,187]	16.314	BRANCH	BRANCH
97	[384,177]	[402,211]	40.456	BRANCH	BRANCH
98	[432,184]	[436,189]	5.657	BRANCH	BRANCH
99	[369,187]	[366,211]	24.243	BRANCH	BRANCH
100	[436,189]	[442,189]	5.0	BRANCH	BRANCH
101	[436,189]	[410,211]	33.698	BRANCH	BRANCH
102	[442,189]	[448,226]	39.314	BRANCH	BRANCH
103	[333,207]	[299,229]	41.698	BRANCH	BRANCH
104	[333,207]	[366,211]	33.243	BRANCH	BRANCH
105	[7,193]	[9,210]	16.828	TIP	BRANCH
106	[0,225]	[9,210]	18.314	TIP	BRANCH
107	[9,210]	[27,246]	43.213	BRANCH	BRANCH
108	[366,211]	[372,219]	10.071	BRANCH	BRANCH
109	[402,211]	[410,211]	7.0	BRANCH	BRANCH
110	[402,211]	[372,219]	31.899	BRANCH	BRANCH
111	[410,211]	[419,222]	14.314	BRANCH	BRANCH
112	[510,228]	[495,214]	19.799	TIP	BRANCH
113	[495,214]	[481,224]	18.314	BRANCH	BRANCH
114	[372,219]	[367,244]	26.071	BRANCH	BRANCH
115	[425,218]	[419,222]	6.657	TIP	BRANCH
116	[419,222]	[420,238]	15.414	BRANCH	BRANCH
117	[481,224]	[463,229]	19.071	BRANCH	BRANCH
118	[481,224]	[491,248]	26.728	BRANCH	BRANCH
119	[448,226]	[463,229]	15.243	BRANCH	BRANCH
120	[448,226]	[427,238]	24.556	BRANCH	BRANCH
121	[299,229]	[295,239]	10.657	BRANCH	BRANCH
122	[269,201]	[299,229]	40.77	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[463,229]	[463,245]	15.0	BRANCH	BRANCH
124	[420,238]	[427,238]	6.0	BRANCH	BRANCH
125	[420,238]	[400,260]	30.627	BRANCH	BRANCH
126	[427,238]	[449,262]	35.042	BRANCH	BRANCH
127	[290,239]	[295,239]	4.0	BRANCH	BRANCH
128	[241,219]	[290,239]	55.87	TIP	BRANCH
129	[273,250]	[290,239]	20.142	TIP	BRANCH
130	[295,239]	[300,265]	26.657	BRANCH	BRANCH
131	[367,244]	[321,265]	54.87	BRANCH	BRANCH
132	[367,244]	[371,270]	26.657	BRANCH	BRANCH
133	[463,245]	[449,262]	21.971	BRANCH	BRANCH
134	[463,245]	[491,248]	31.142	BRANCH	BRANCH
135	[33,245]	[27,246]	5.414	TIP	BRANCH
136	[27,246]	[35,275]	31.314	BRANCH	BRANCH
137	[153,246]	[150,248]	2.828	TIP	BRANCH
138	[106,270]	[150,248]	61.87	TIP	BRANCH
139	[145,244]	[150,248]	5.243	TIP	BRANCH
140	[491,248]	[490,287]	57.385	BRANCH	BRANCH
141	[400,260]	[371,270]	32.142	BRANCH	BRANCH
142	[400,260]	[407,284]	25.485	BRANCH	BRANCH
143	[449,262]	[449,268]	5.0	BRANCH	BRANCH
144	[300,265]	[321,265]	20.0	BRANCH	BRANCH
145	[300,265]	[280,282]	26.799	BRANCH	BRANCH
146	[321,265]	[334,283]	22.556	BRANCH	BRANCH
147	[449,268]	[435,285]	22.385	BRANCH	BRANCH
148	[449,268]	[490,287]	47.456	BRANCH	BRANCH
149	[371,270]	[343,304]	52.799	BRANCH	BRANCH
150	[75,248]	[35,275]	51.941	TIP	BRANCH
151	[30,307]	[35,275]	33.071	TIP	BRANCH
152	[221,265]	[233,275]	24.899	TIP	BRANCH
153	[199,277]	[233,275]	34.657	TIP	BRANCH
154	[241,289]	[233,275]	16.728	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[276,273]	[280,282]	9.243	TIP	BRANCH
156	[280,282]	[261,307]	33.213	BRANCH	BRANCH
157	[334,283]	[343,304]	25.485	BRANCH	BRANCH
158	[334,283]	[316,314]	40.456	BRANCH	BRANCH
159	[407,284]	[435,285]	27.414	BRANCH	BRANCH
160	[407,284]	[397,313]	32.314	BRANCH	BRANCH
161	[435,285]	[440,310]	26.071	BRANCH	BRANCH
162	[490,287]	[467,321]	42.698	BRANCH	BRANCH
163	[78,291]	[77,304]	12.414	TIP	BRANCH
164	[77,304]	[100,306]	22.828	BRANCH	BRANCH
165	[77,304]	[41,347]	61.012	BRANCH	BRANCH
166	[343,304]	[316,314]	30.728	BRANCH	BRANCH
167	[101,303]	[100,306]	2.414	TIP	BRANCH
168	[238,314]	[261,307]	24.899	TIP	BRANCH
169	[261,307]	[316,314]	62.284	BRANCH	BRANCH
170	[440,310]	[435,321]	12.657	BRANCH	BRANCH
171	[440,310]	[467,321]	30.142	BRANCH	BRANCH
172	[397,313]	[418,332]	29.627	BRANCH	BRANCH
173	[397,313]	[385,327]	20.314	BRANCH	BRANCH
174	[208,299]	[212,318]	19.657	TIP	BRANCH
175	[224,319]	[212,318]	12.243	TIP	BRANCH
176	[203,322]	[212,318]	9.243	TIP	BRANCH
177	[435,321]	[418,332]	20.556	BRANCH	BRANCH
178	[435,321]	[467,328]	35.142	BRANCH	BRANCH
179	[467,321]	[467,328]	6.0	BRANCH	BRANCH
180	[385,327]	[363,341]	30.899	BRANCH	BRANCH
181	[385,327]	[403,342]	23.799	BRANCH	BRANCH
182	[467,328]	[484,343]	21.799	BRANCH	BRANCH
183	[418,332]	[418,334]	1.0	BRANCH	BRANCH
184	[418,334]	[403,342]	17.314	BRANCH	BRANCH
185	[418,334]	[443,353]	32.627	BRANCH	BRANCH
186	[363,341]	[334,342]	30.071	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[363,341]	[367,353]	12.243	BRANCH	BRANCH
188	[334,342]	[309,357]	31.971	BRANCH	BRANCH
189	[334,342]	[327,360]	20.485	BRANCH	BRANCH
190	[403,342]	[367,353]	41.142	BRANCH	BRANCH
191	[484,343]	[491,343]	6.0	BRANCH	BRANCH
192	[484,343]	[482,384]	42.485	BRANCH	BRANCH
193	[469,355]	[484,343]	18.556	TIP	BRANCH
194	[492,320]	[491,343]	22.414	TIP	BRANCH
195	[491,343]	[496,344]	4.0	BRANCH	BRANCH
196	[502,342]	[496,344]	5.414	TIP	BRANCH
197	[502,358]	[496,344]	15.071	TIP	BRANCH
198	[1,362]	[41,347]	47.213	TIP	BRANCH
199	[41,347]	[47,368]	22.071	BRANCH	BRANCH
200	[367,353]	[367,367]	13.0	BRANCH	BRANCH
201	[454,346]	[443,353]	12.899	TIP	BRANCH
202	[443,353]	[461,391]	44.456	BRANCH	BRANCH
203	[257,333]	[242,357]	30.971	TIP	BRANCH
204	[242,357]	[223,386]	36.627	BRANCH	BRANCH
205	[242,357]	[264,364]	23.485	BRANCH	BRANCH
206	[309,357]	[264,364]	47.728	BRANCH	BRANCH
207	[309,357]	[327,360]	17.828	BRANCH	BRANCH
208	[327,360]	[326,376]	15.414	BRANCH	BRANCH
209	[260,377]	[264,364]	13.657	TIP	BRANCH
210	[367,367]	[356,378]	14.728	BRANCH	BRANCH
211	[384,374]	[367,367]	18.485	TIP	BRANCH
212	[64,369]	[47,368]	16.414	TIP	BRANCH
213	[44,412]	[47,368]	49.213	TIP	BRANCH
214	[326,376]	[356,378]	30.657	BRANCH	BRANCH
215	[326,376]	[312,398]	26.385	BRANCH	BRANCH
216	[355,380]	[356,378]	1.414	TIP	BRANCH
217	[193,368]	[195,379]	10.828	TIP	BRANCH
218	[195,379]	[223,386]	29.899	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[190,383]	[195,379]	5.243	TIP	BRANCH
220	[482,384]	[496,401]	23.556	BRANCH	BRANCH
221	[482,384]	[466,391]	17.899	BRANCH	BRANCH
222	[223,386]	[278,433]	100.456	BRANCH	BRANCH
223	[461,391]	[466,391]	4.0	BRANCH	BRANCH
224	[461,391]	[441,418]	34.456	BRANCH	BRANCH
225	[466,391]	[482,411]	27.971	BRANCH	BRANCH
226	[317,404]	[312,398]	6.657	TIP	BRANCH
227	[496,401]	[482,411]	17.142	BRANCH	BRANCH
228	[509,417]	[496,401]	19.971	TIP	BRANCH
229	[376,407]	[379,422]	15.243	BRANCH	BRANCH
230	[337,400]	[376,407]	40.485	TIP	BRANCH
231	[376,407]	[416,412]	77.184	BRANCH	BRANCH
232	[482,411]	[485,445]	40.042	BRANCH	BRANCH
233	[416,412]	[391,423]	28.556	BRANCH	BRANCH
234	[416,412]	[441,418]	26.071	BRANCH	BRANCH
235	[441,418]	[441,420]	1.0	BRANCH	BRANCH
236	[441,420]	[478,445]	46.941	BRANCH	BRANCH
237	[443,434]	[441,420]	13.828	TIP	BRANCH
238	[379,422]	[391,423]	11.414	BRANCH	BRANCH
239	[379,422]	[341,430]	39.899	BRANCH	BRANCH
240	[391,423]	[391,440]	16.0	BRANCH	BRANCH
241	[341,430]	[308,433]	33.243	BRANCH	BRANCH
242	[341,430]	[354,473]	47.385	BRANCH	BRANCH
243	[281,386]	[278,433]	57.042	TIP	BRANCH
244	[278,433]	[308,433]	29.0	BRANCH	BRANCH
245	[297,464]	[308,433]	34.556	TIP	BRANCH
246	[391,440]	[383,455]	17.485	BRANCH	BRANCH
247	[420,451]	[391,440]	32.142	TIP	BRANCH
248	[478,445]	[485,445]	6.0	BRANCH	BRANCH
249	[471,462]	[478,445]	18.485	TIP	BRANCH
250	[509,456]	[485,445]	28.314	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
251	[383,455]	[382,474]	19.243	BRANCH	BRANCH
252	[369,449]	[383,455]	15.071	TIP	BRANCH
253	[433,466]	[434,471]	4.414	TIP	BRANCH
254	[452,487]	[434,471]	24.799	TIP	BRANCH
255	[434,471]	[395,479]	41.728	BRANCH	BRANCH
256	[354,473]	[382,474]	27.828	BRANCH	BRANCH
257	[354,499]	[354,473]	27.071	TIP	BRANCH
258	[382,474]	[395,479]	14.657	BRANCH	BRANCH
259	[395,479]	[399,495]	16.657	BRANCH	BRANCH
260	[34,470]	[35,482]	11.414	TIP	BRANCH
261	[35,482]	[22,484]	12.828	BRANCH	BRANCH
262	[66,493]	[35,482]	34.142	TIP	BRANCH
263	[0,464]	[22,484]	32.213	TIP	BRANCH
264	[9,509]	[22,484]	29.971	TIP	BRANCH
265	[498,477]	[498,485]	7.0	TIP	BRANCH
266	[452,509]	[498,485]	56.698	TIP	BRANCH
267	[506,488]	[498,485]	7.828	TIP	BRANCH
268	[394,497]	[399,495]	4.828	TIP	BRANCH
269	[191,63]	[160,78]	39.213	TIP	TIP
270	[87,106]	[94,110]	8.657	TIP	TIP
271	[203,123]	[184,136]	25.556	TIP	TIP
272	[23,145]	[22,157]	14.071	TIP	TIP
273	[160,154]	[173,161]	15.899	TIP	TIP
274	[252,177]	[237,183]	17.485	TIP	TIP
275	[219,194]	[197,220]	39.698	TIP	TIP
276	[102,198]	[110,216]	21.314	TIP	TIP
277	[55,199]	[64,214]	19.314	TIP	TIP
278	[171,203]	[161,227]	28.728	TIP	TIP
279	[200,224]	[206,235]	13.485	TIP	TIP
280	[116,225]	[138,242]	30.799	TIP	TIP
281	[177,279]	[168,291]	16.314	TIP	TIP
282	[64,284]	[62,293]	9.828	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
283	[139,305]	[117,312]	25.728	TIP	TIP
284	[510,313]	[503,317]	8.657	TIP	TIP
285	[7,314]	[2,315]	5.414	TIP	TIP
286	[182,326]	[121,351]	74.426	TIP	TIP
287	[103,332]	[103,351]	20.657	TIP	TIP
288	[216,340]	[200,349]	20.314	TIP	TIP
289	[175,404]	[182,407]	8.243	TIP	TIP
290	[13,421]	[20,441]	23.485	TIP	TIP
291	[78,430]	[84,440]	12.485	TIP	TIP
292	[267,466]	[266,467]	1.414	TIP	TIP
293	[236,475]	[229,476]	8.243	TIP	TIP
294	[258,494]	[274,498]	18.485	TIP	TIP
295	[412,502]	[421,511]	14.485	TIP	TIP